

Homework 6

Department :

Computer Science and Engineering

Invigilator's Signature

Total No. of Pages

Mid Semester

End Semester

Make - up

Suppl. Exam

STUDENT'S NAME

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Roll No. :

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Programme :

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Course No. :

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Course Title :

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PART :

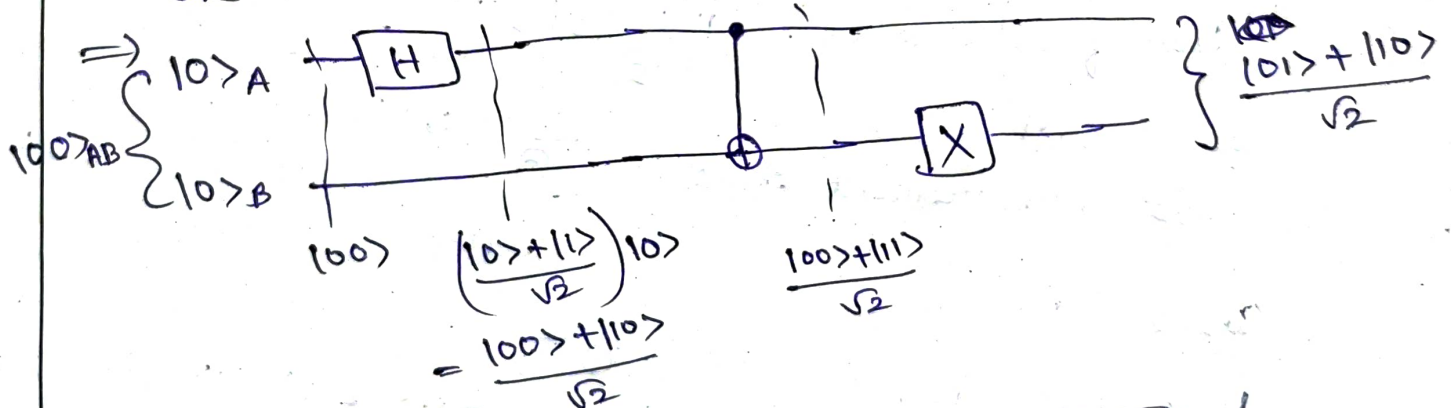
Question No.	1	2	3	4	5	6	7	8	9	10	Total
MARKS											

Homework 6

1. (a) $\Rightarrow$ Initial joint state of AB = $|0\rangle_A \otimes |0\rangle_B = |00\rangle_{AB}$

$\Rightarrow$ Find unitary matrix (gate U) such that
 $U|00\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$; $UU^\dagger = U^\dagger U = I$

$\Rightarrow A \in C^2 = C^{d_1}$, $B \in C^2 = C^{d_2}$, $d_1 = d_2 = 2$



$\Rightarrow U = (I_{d_1} \otimes X)(CNOT_{A \rightarrow B})(H \otimes I_{d_2})$

$$\Rightarrow U = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \quad \text{in matrix form}$$

(b) $\Rightarrow$ Suppose we measure observable Z on qubit A (local measurement). What are possible post measurement states of AB and with what probabilities do they occur?

$$\Rightarrow \text{Given state } |\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

Observable Z on qubit A.

$$Z|0\rangle = +1|0\rangle, \quad Z|1\rangle = -1|1\rangle$$

Outcome = +1 $\rightarrow$ qubit A in $|0\rangle$

Outcome = -1 $\rightarrow$ qubit A in $|1\rangle$

Case 1: Outcome = +1

Projector for this outcome = $P_{+1} = |0\rangle\langle 0| \otimes I_2$

$$P_{+1}|\psi\rangle = (|0\rangle\langle 0| \otimes I_2) \left(\frac{1}{\sqrt{2}} |01\rangle + \frac{1}{\sqrt{2}} |10\rangle \right)$$

$$= \frac{1}{\sqrt{2}} |0\rangle\langle 0| \otimes |1\rangle + 0 = \frac{1}{\sqrt{2}} |01\rangle$$

$$\text{Prob. of outcome +1} = \left\| \frac{1}{\sqrt{2}} |01\rangle \right\|^2 = \frac{1}{2} \rightarrow$$

~~Case 2: Outcome = -1~~

$$\text{Post measurement state} = \frac{\frac{1}{\sqrt{2}} |01\rangle}{\sqrt{\text{prob.}}} = \frac{\frac{1}{\sqrt{2}} |01\rangle}{\sqrt{\frac{1}{2}}} = |01\rangle$$

Case 2: Outcome = -1

Projector = $P_{-1} = |1\rangle\langle 1| \otimes I_2$

$$P_{-1}|\psi\rangle = (|1\rangle\langle 1| \otimes I_2) \left(\frac{1}{\sqrt{2}} |01\rangle + \frac{1}{\sqrt{2}} |10\rangle \right)$$

$$= 0 + \frac{1}{\sqrt{2}} |1\rangle\langle 1| \otimes |0\rangle = \frac{1}{\sqrt{2}} |10\rangle$$

$$\text{Probability of outcome -1} = \left\| \frac{1}{\sqrt{2}} |10\rangle \right\|^2 = \frac{1}{2}$$

Post measurement state = $\frac{\frac{1}{\sqrt{2}}|10\rangle}{\sqrt{\text{Prob.}}} = \frac{\frac{1}{\sqrt{2}}|10\rangle}{\sqrt{\frac{1}{2}}} = |10\rangle$

2.

$\Rightarrow \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ — computational basis

$\Rightarrow$ Choose following 4 orthonormal entangled states:

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\psi_2\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$$

$$|\psi_3\rangle = \frac{1}{\sqrt{6}}(|00\rangle - 2|10\rangle - |11\rangle)$$

$$|\psi_4\rangle = \frac{1}{2\sqrt{3}}(|00\rangle - 3|01\rangle + |10\rangle - |11\rangle)$$

(a) $\Rightarrow$ Define projection operators.

$$P_i = |\psi_i\rangle\langle\psi_i| \quad i = 1, 2, 3, 4$$

These are orthogonal projectors; $\{|\psi_i\rangle\}$ is orthonormal basis. $\langle\psi_i|\psi_j\rangle = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$

$$P_i P_j = \delta_{ij} P_i \quad ; \quad \sum_{i=1}^4 P_i = I$$

Define observable

$$M = P_1 + 2P_2 + 3P_3 + 4P_4$$

P_i 's are orthogonal projectors $\rightarrow M$ is Hermitian

Eigenvalues of M are 1, 2, 3, 4 (distinct) with eigenvectors $|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, |\psi_4\rangle$. Projection operators corresponding to distinct eigenvalues are P_1, P_2, P_3, P_4 .

(b) $\Rightarrow$ Measurement from initial state $|00\rangle$

$$\Rightarrow |00\rangle = \frac{1}{\sqrt{2}}|\psi_1\rangle + \frac{1}{2}|\psi_2\rangle + \frac{1}{\sqrt{6}}|\psi_3\rangle + \frac{1}{2\sqrt{3}}|\psi_4\rangle$$

$\Rightarrow$ Measuring observable M , outcome i occurs with probability $P_i = |\langle\psi_i|00\rangle|^2$

$$P_1 = \frac{1}{2}, P_2 = \frac{1}{4}, P_3 = \frac{1}{6}, P_4 = \frac{1}{12}$$

Post measurement state for outcome i is $|\psi_i\rangle$
for $i=1, 2, 3, 4$.

Proof that every post measurement state is entangled

For pure 2-qubit state,

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

State is entangled if $\alpha\delta - \beta\gamma \neq 0$.

[For product state, $|\psi\rangle = (m|0\rangle + n|1\rangle) \otimes (p|0\rangle + q|1\rangle)$
and $\alpha\delta = \beta\gamma$]

$$\textcircled{1} |\psi_1\rangle \rightarrow \alpha\delta - \beta\gamma = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - 0 \cdot 0 = \frac{1}{2} \neq 0$$

$|\psi_1\rangle$ is entangled state

$$\textcircled{2} |\psi_2\rangle \rightarrow \alpha\delta - \beta\gamma = \left(\frac{1}{2}\right)\left(\frac{-1}{2}\right) - \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = -\frac{1}{2} \neq 0$$

$|\psi_2\rangle$ is entangled state

$$\textcircled{3} |\psi_3\rangle \rightarrow \alpha\delta - \beta\gamma = \left(\frac{1}{\sqrt{6}}\right)\left(\frac{-1}{\sqrt{6}}\right) - (0)\left(\frac{-2}{\sqrt{6}}\right) = \frac{-1}{6} \neq 0$$

$|\psi_3\rangle$ is entangled state

$$\textcircled{4} |\psi_4\rangle \rightarrow \alpha\delta - \beta\gamma = \left(\frac{1}{2\sqrt{3}}\right)\left(\frac{-1}{2\sqrt{3}}\right) - \left(\frac{-3}{2\sqrt{3}}\right)\left(\frac{1}{2\sqrt{3}}\right) = \frac{1}{6} \neq 0$$

$|\psi_4\rangle$ is entangled state

3.

$$(a) \Rightarrow \boxed{CNOT_{A \rightarrow B} = U}$$

$$\Rightarrow \cancel{U|00\rangle = |00\rangle}, U|0\rangle_A |0\rangle_B = |0\rangle_A |0\rangle_B$$

$$\cancel{U|0\rangle_A |1\rangle_B = |0\rangle_A |0\rangle_B}$$

$$U|1\rangle_A |0\rangle_B = |1\rangle_A |1\rangle_B$$

State of
B = state of
A

$\Rightarrow U$ copies state of A into state of B. (CNOT gate)

$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

State of B
initially = $|0\rangle_B$

State of A restricted to $\{|0\rangle, |1\rangle\}$ only

$$(b) \Rightarrow CNOT_{A \rightarrow B} (|+\rangle_A |0\rangle_B) = \cancel{CNOT_{A \rightarrow B}} \left[|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right]$$

$$= \frac{1}{\sqrt{2}} CNOT_{A \rightarrow B} |0\rangle_A |0\rangle_B + \frac{1}{\sqrt{2}} CNOT_{A \rightarrow B} |1\rangle_A |0\rangle_B$$

$$= \frac{1}{\sqrt{2}} |0\rangle_A |0\rangle_B + \frac{1}{\sqrt{2}} |1\rangle_A |1\rangle_B = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$\Rightarrow \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$ is entangled state, not a product state.

It won't clone $|+\rangle$ into B.

$$|+\rangle_A |+\rangle_B = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

B is not clone of A

(c) $\Rightarrow$ Make U such that orthonormal set is $\{|+\rangle, |-\rangle\}$

$\Rightarrow H =$ Hadamard gate.

$$H|0\rangle = |+\rangle, H|1\rangle = |-\rangle$$

$$H|+\rangle = |0\rangle, H|-\rangle = |1\rangle$$

$$\Rightarrow \boxed{U = (H \otimes H) CNOT_{A \rightarrow B} (H \otimes I)}$$

$$\begin{aligned}
 &\Rightarrow U (|+\rangle_A |0\rangle_B) \quad [\text{State of B initially} \\
 &\quad \quad \quad = |0\rangle] \\
 &= (H \otimes H) \text{CNOT}_{A \rightarrow B} (H \otimes I) (|+\rangle_A |0\rangle_B) \\
 &= (H \otimes H) \text{CNOT}_{A \rightarrow B} (H \otimes I) \left(\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle \right) \\
 &= (H \otimes H) \text{CNOT}_{A \rightarrow B} \left(\frac{1}{\sqrt{2}} |+\rangle |0\rangle + \frac{1}{\sqrt{2}} |-\rangle |0\rangle \right) \\
 &= (H \otimes H) \text{CNOT}_{A \rightarrow B} \left(\frac{1}{2} |00\rangle + \frac{1}{2} |10\rangle + \frac{1}{2} |00\rangle - \frac{1}{2} |10\rangle \right) \\
 &= (H \otimes H) \text{CNOT}_{A \rightarrow B} (|00\rangle) \\
 &= (H \otimes H) (|+\rangle |+\rangle) = (|00\rangle) \\
 &= |+\rangle_A |+\rangle_B \quad \longrightarrow \text{A is cloned into B}
 \end{aligned}$$

$$\begin{aligned}
 &\Rightarrow U |-\rangle_A |0\rangle_B \\
 &= (H \otimes H) \text{CNOT}_{A \rightarrow B} (H \otimes I) (|-\rangle_A |0\rangle_B) \\
 &= (H \otimes H) \text{CNOT}_{A \rightarrow B} (H \otimes I) \left(\frac{1}{\sqrt{2}} |00\rangle - \frac{1}{\sqrt{2}} |10\rangle \right) \\
 &= (H \otimes H) \text{CNOT}_{A \rightarrow B} \left(\frac{1}{\sqrt{2}} |+\rangle |0\rangle - \frac{1}{\sqrt{2}} |-\rangle |0\rangle \right) \\
 &= (H \otimes H) \text{CNOT}_{A \rightarrow B} \left(\frac{1}{2} |00\rangle + \frac{1}{2} |10\rangle - \frac{1}{2} |00\rangle + \frac{1}{2} |10\rangle \right) \\
 &= (H \otimes H) \text{CNOT}_{A \rightarrow B} (|10\rangle) = (H \otimes H) (|11\rangle) \\
 &= |-\rangle_A |-\rangle_B \quad \longrightarrow \text{A is cloned into B}
 \end{aligned}$$

4. $\Rightarrow M = \text{measurable}$
 Eigenvalues of $M = \lambda_1, \lambda_2, \dots, \lambda_L$
 Corresponding orthogonal projection operators $P_1, \dots, P_L$.

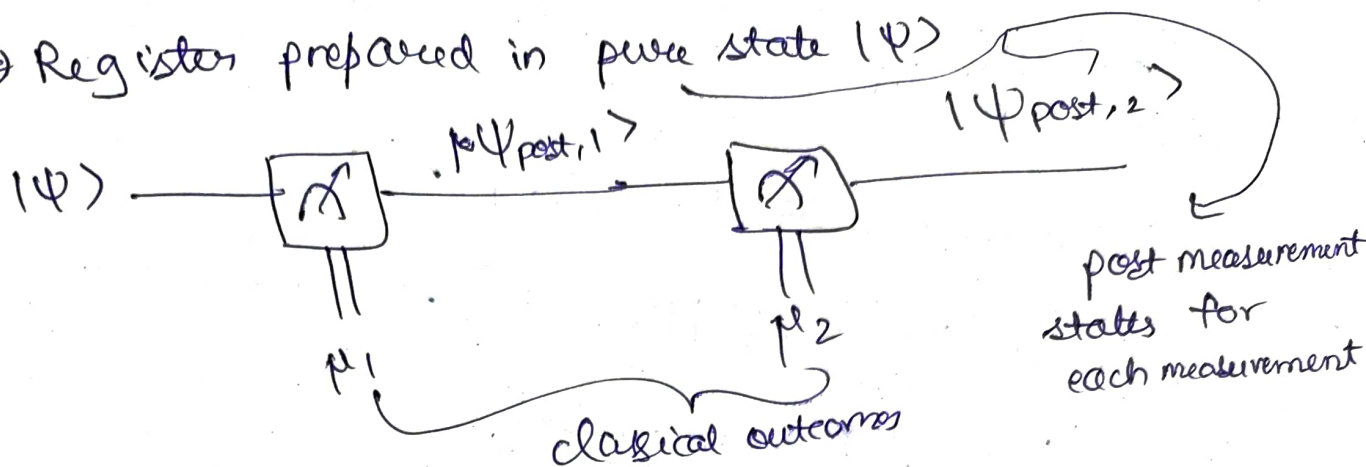
$$P_i P_j = \delta_{ij} P_i \quad ; \quad \sum_{i=1}^L P_i = I \quad \rightarrow \text{completeness condition}$$

$$M = \sum_{i=1}^L \lambda_i P_i$$

$$P_i^2 = P_i^\dagger = P_i \quad \forall i = 1 \text{ to } L$$

$$P_i P_j = 0 \quad \forall i \neq j \quad (\text{mutually orthogonal})$$

⇒ Register prepared in pure state $|\psi\rangle$



⇒ For pure state $|\psi\rangle$,

(a)

$$p_i = \langle \psi | P_i | \psi \rangle \quad ; \quad |\psi_{\text{post},i}\rangle = \frac{P_i |\psi\rangle}{\sqrt{p_i}}$$

⇒ For mixed state (ρ = density operator),

$$p_i = \text{Tr}[P_i \rho P_i^\dagger] \quad ; \quad \rho_{\text{post},i} = \frac{P_i \rho P_i^\dagger}{\text{Tr}[P_i \rho P_i^\dagger]}$$

(a)

⇒ let's take some pure state $|\psi\rangle$ initially

⇒ let 1st measurement gives outcome λ_i

$$p_i = \langle \psi | P_i | \psi \rangle \quad ; \quad |\psi_{\text{post},i}\rangle = \frac{P_i |\psi\rangle}{\sqrt{p_i}}$$

Measure same observable again now. Given 1st outcome is λ_i , prob. to obtain λ_j on 2nd measurement

$$\text{Pr}(M_2 = \lambda_j | M_1 = \lambda_i) = \langle \psi_{\text{post},i} | P_j | \psi_{\text{post},i} \rangle$$

$$|\psi_{\text{post},i}\rangle = \frac{P_i |\psi\rangle}{\sqrt{p_i}}$$

$$\text{Pr}(M_2 = \lambda_j | M_1 = \lambda_i) = \frac{1}{p_i} \langle \psi | P_i P_j P_i | \psi \rangle$$

$$= \frac{\delta_{ij}}{p_i} \langle \psi | P_i | \psi \rangle$$

$$= \frac{\delta_{ij}}{p_i} \cdot p_i = \delta_{ij}$$

$$\Rightarrow \Pr(M_2 = \lambda_i | M_1 = \lambda_i) = 1$$

$$\Pr(M_2 = \lambda_j | M_1 = \lambda_i) = 0 \text{ if } i \neq j$$

$$\boxed{M_2 = M_1 \text{ with prob. } 1}$$

$\Rightarrow$ Post measurement state after 2nd measurement

$$= |\psi_{\text{post},2}\rangle = \frac{P_i |\psi_{\text{post},1}\rangle}{\sqrt{\langle \psi_{\text{post},1} | P_i | \psi_{\text{post},1} \rangle}}$$

$$= \frac{P_i \frac{P_i |\psi\rangle}{\sqrt{p_i}}}{1} = \frac{P_i^2 |\psi\rangle}{\sqrt{p_i}} = \frac{P_i |\psi\rangle}{\sqrt{p_i}}$$

$$\boxed{|\psi_{\text{post},2}\rangle = |\psi_{\text{post},1}\rangle}$$

(b) $\Rightarrow$ Let state be fed into 1st measurement is an arbitrary (possibly mixed state) ρ .

$\Rightarrow$ Let initial state = $\rho \longrightarrow$ some arbitrary mixed state.

$\Rightarrow$ 1st measurement gives outcome λ_i with prob. $p_i = \text{Tr}(P_i \rho P_i^\dagger) = \text{Tr}(P_i \rho)$

Post measurement state = ρ_i

$$= \frac{P_i \rho P_i^\dagger}{\text{Tr}[P_i \rho P_i^\dagger]}$$

⇒ Measure same observable 2nd time. Given 1st outcome is λ_i , prob. of obtaining λ_j on 2nd measurement is

$$Pr(M_2 = \lambda_j | M_1 = \lambda_i) = \text{Tr}[P_j S_1 P_j^\dagger]$$

$$= \text{Tr}\left[P_j \frac{P_i S_1 P_i^\dagger}{\text{Tr}[P_i S_1 P_i^\dagger]} P_j^\dagger\right]$$

$$= \frac{\text{Tr}[(P_j P_i) S_1 (P_j P_i)^\dagger]}{\text{Tr}[P_i S_1 P_i^\dagger]}$$

$$= \frac{\delta_{ij} \phi_i}{\phi_i} = \delta_{ij}$$

$$P_i P_j = \delta_{ij} P_i$$

$$Pr(M_2 = \lambda_i | M_1 = \lambda_i) = 1$$

$$Pr(M_2 = \lambda_j | M_1 = \lambda_i) = 0 \quad \forall i \neq j$$

$$\boxed{M_2 = M_1 \text{ with prob. } 1}$$

⇒ Post measurement state after 2nd measurement

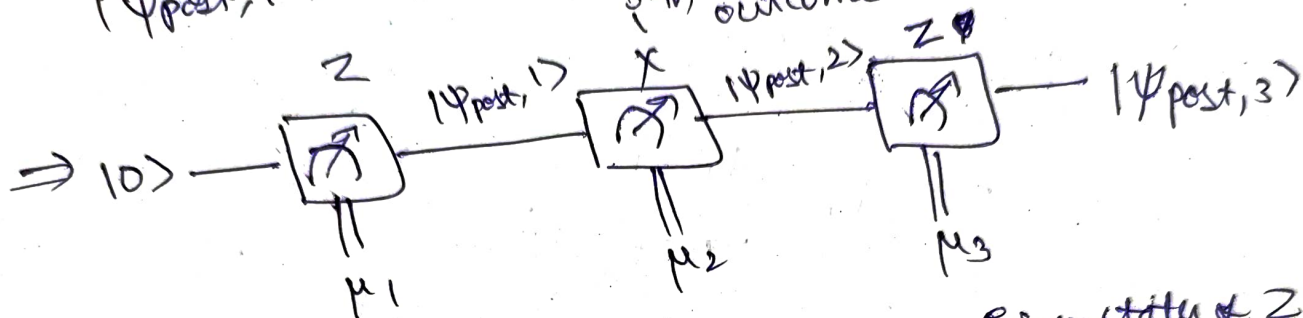
$$= S_2 = \frac{P_i S_1 P_i^\dagger}{\text{Tr}[P_i S_1 P_i^\dagger]} = P_i S_1 P_i^\dagger$$

$$= P_i \left(\frac{P_i S_1 P_i^\dagger}{\phi_i} \right) P_i^\dagger = \frac{P_i S_1 P_i}{\phi_i} = S_1$$

$$\boxed{S_2 = S_1}$$

5. $\Rightarrow$ Initial state of qubit (registers) = $|0\rangle$
 μ_i — classical outcome

$|\psi_{\text{post}, i}\rangle$ — post measurement state of i -th outcome



$\Rightarrow Z|0\rangle = |0\rangle$; $Z|1\rangle = -|1\rangle \rightarrow$ Eigenstates of Z

$X|+\rangle = |+\rangle$; $X|-\rangle = -|-\rangle \rightarrow$ Eigenstates of X

Measure Z then X then Z

$\Rightarrow$ 1st measurement: Initial state = $|0\rangle$
 $|0\rangle \rightarrow$ eigenvector (eigenstate) of Z with eigenvalue 1

$$\boxed{\mu_1 = +1}$$

$$\boxed{|\psi_{\text{post}, 1}\rangle = |0\rangle}$$

$\Rightarrow$ 2nd measurement: $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$

$$\boxed{\begin{array}{l} \mu_2 = +1 \text{ with prob. } \frac{1}{2}, |\psi_{\text{post}, 2}\rangle = |+\rangle \\ \mu_2 = -1 \text{ with prob. } \frac{1}{2}, |\psi_{\text{post}, 2}\rangle = |-\rangle \end{array}}$$

$\Rightarrow$ 3rd measurement: $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

① If state = $|+\rangle$ after X measurement,

$$\mu_3 = 1, |\psi_{\text{post}, 3}\rangle = |0\rangle$$

$$\mu_3 = -1, |\psi_{\text{post}, 3}\rangle = |1\rangle$$

② If state is $1 \rightarrow$ after X measurement,
 $\mu_3 = +1$ with prob. $\frac{1}{2}$, $|\psi_{\text{post},3}\rangle = |0\rangle$
 $\mu_3 = -1$ with prob. $\frac{1}{2}$, $|\psi_{\text{post},3}\rangle = |1\rangle$

$\Rightarrow$ Regardless of X -measurement outcome,

$$P(\mu_3 = -1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

$\Rightarrow \mu_1 \neq \mu_3$ with $\mu_1 = +1$ ~~and~~ always
 $\mu_3 = -1$ with prob. $\frac{1}{2}$ and $\mu_3 = +1$ with prob. $\frac{1}{2}$

$$\boxed{P(\mu_1 \neq \mu_3) = \frac{1}{2}} \rightarrow \text{non-zero probability}$$

$\Rightarrow |\psi_{\text{post},1}\rangle = |0\rangle$ always

$$P(|\psi_{\text{post},3}\rangle = |01\rangle) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

$|\psi_{\text{post},3}\rangle = |0\rangle$ with prob. $\frac{1}{2}$ and $|1\rangle$ with prob. $\frac{1}{2}$

$$\boxed{P(|\psi_{\text{post},3}\rangle \neq |\psi_{\text{post},1}\rangle) = \frac{1}{2}}$$

$\hookrightarrow$ non zero probability

THE END